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Gas Turbines for Electric Power Generation Coupled with Battery Energy Storage Systems and Renewable Energy Sources to Decarbonize LNG Applications

G. Betti, D. Sgrò, R. Palmieri, and D. Nurzad, Baker Hughes, Florence, Italy; S. Grifa, Baker Hughes - Nuovo Pignone International, Florence, Italy

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Abstract

Due to its ease of transport and relatively low carbon dioxide emissions, Liquefied Natural Gas (LNG) represents an interesting alternative to other fossil fuels. In the next future, it is expected that the LNG demand will steadily increase and that new LNG plants will be constructed across the globe. LNG plants are often located in remote areas without a connection to the electrical grid and are therefore equipped with off-grid captive power generation plants having a high level of availability, usually based on Open Cycle Gas Turbines (OCGT) which are a source of CO₂ emissions. Methods and solutions to decarbonize LNG applications will be presented and analyzed.

An option to partially decarbonize the LNG facilities consists in adopting Combined Cycle Gas Turbines (CCGT), characterized by lower fuel consumptions with respect to OCGT. In addition, the integration of Battery Energy Storage Systems (BESS) can provide spinning reserve, thus allowing OCGT and CCGT to run at higher partial loads having higher efficiency. A further step to decrease CO₂ emissions is constituted by the adoption of Renewable Energy Sources (RES), however characterized by fluctuations in the output power, that represent a risk for plant availability. Each one of the listed technologies, CCGT, BESS and RES, leads to an increase in the Capital Expenditure (CAPEX) but, if properly sized, results in decreasing operating expenses (OPEX) and greenhouse gases emissions. In this work, for each use case, the optimal plant configuration and the optimal hourly scheduling of the power sources are computed via advanced optimization algorithms. These algorithms guarantee the minimization of the LCOE, while fulfilling both the electrical/thermal loads, the constraints on plant availability and the operational constraints of each piece of equipment.

This paper presents a comprehensive analysis highlighting the contribution of each one of the evaluated technologies to the CAPEX, OPEX, CO₂ emissions and Levelized Cost of Energy (LCOE) of a selected LNG plant. CCGT, BESS and RES are introduced via subsequent steps in the technology mix of the captive power island, so that it is possible to clearly define the benefits generated by each one of them, and the relative boundaries of technical and economic feasibility considering current market prices.

The optimal design of hybrid power plants for electrified LNG facilities constitutes a novel approach towards an economically feasible decarbonization path for such kind of plants. The clear explanation of the contribution provided by each installed technology is key to understand the benefits of each portion of the initial investment. The robustness of the proposed solutions is guaranteed by strategies to mitigate the fast fluctuations of photovoltaic (PV) plants, that are shown and considered in the costs of the PV generators.

Introduction

An LNG facility treats and cools raw natural gas until it condenses, shrinking its volume to just 1/600th of its original size. This operation demands significant energy: the plant must operate non-stop and relies heavily on a stable power supply ([Lim 2013](#)). In remote areas, many LNG operators opt to generate their own electricity through captive power plants, using a portion of the processed gas to fuel the generators. For electrified LNG facilities (e-LNG), which require large electric motor-driven compressors, the captive power plant capacity can easily reach several hundred megawatts ([Gambini 2023](#)).

It is widely recognized that generating electricity from fossil fuels—including that used in e-LNG plants—is a major contributor to human-induced greenhouse gas (GHG) emissions ([Bruschini 2022](#)). This paper focuses on a mid-scale e-LNG plant and explores the decarbonization potential of integrating a utility-scale Battery Energy Storage System (BESS) and a photovoltaic (PV) plant, combined with a balanced mix of Simple Cycle and Combined Cycle Gas Turbines (SCGT/CCGT) ([Poullikkas 2005](#)), forming a hybrid power system.

Incorporating CCGT into a fully fossil-fueled power setup helps reduce GHG emissions by utilizing gas turbine exhaust to generate steam, which in turn powers a Steam Turbine (ST) for additional electricity. BESS also plays a crucial role in this context: it can be seamlessly added to power plants to provide spinning reserve or to stabilize fluctuations in output from Renewable Energy Sources (RES). Moreover, RES—offering zero-emission electricity—can further improve performance compared to traditional open-cycle Gas Turbine Generators (GTGs) ([Jia 2023](#)). However, integrating intermittent renewable sources presents three main challenges:

- Significant day/night power swings must be smoothed by a properly sized BESS when renewable penetration is high.
- RES inherently produce rapid and frequent power changes, while CCGTs are designed to operate near full load and to avoid frequent cycling.
- In off-grid hybrid systems, the entire generation mix must ensure spinning reserve and high availability to prevent load shedding and operational issues.

This study assesses how combining SCGT, CCGT, BESS, and PV systems influences the performance, cost-effectiveness, and carbon emissions of an LNG plant that requires both electricity and heat. It also evaluates whether such a hybrid setup can match the reliability of traditional systems while offering environmental and economic benefits. The fast output power fluctuations of PV plants are taken into account and mitigated by dedicated storage systems.

Considered technologies

When paired with a well-designed captive power generation system, electrifying gas compression trains can significantly contribute to the decarbonization of LNG facilities ([Bruschini 2022](#); [Gambini 2023](#)). Baker Hughes was among the first to implement electric motors as drivers for LNG refrigerant trains, including the compact liquefaction modules (NMBL™) considered as load in this study. They have capacities ranging from 0.8 to 2 MTPA, and they can be added incrementally as fully tested, plug-and-play systems, allowing for capacity expansion while minimizing financial risk. The required power for the electric motor of each module is equal to 42 MW.

On the power generation front, to improve the flexibility and responsiveness of the power island, aeroderivative gas turbines paired with 2-pole synchronous generators are considered among the best options for supplying electricity to motor-driven compression trains (Bruschini 2022). Specifically, this study considers a configuration based on the LM9000 due to its strong technical performance (Corsini 2020; Marchetti 2020; Mazzola 2020; Pelagotti 2019). The Baker Hughes LM9000 is characterized by a simple cycle efficiency exceeding 40%, making it one of the most efficient gas turbines in the 65+ MW class. Its rapid ramp-up capability, moreover, make it especially well-suited for hybrid power systems that incorporate renewable energy. Efficiency gains beyond those of a Simple Cycle Gas Turbine (SCGT) can be achieved by operating in a Combined Cycle Gas Turbine (CCGT) configuration, where exhaust gases from the gas turbine are used to drive a steam turbine generator (Gambini 2023; Poullikkas 2005). The CCGT setup selected by Baker Hughes enhances plant flexibility and ensures that even during open cycle operation—common during various phases of a plant's lifecycle—performance and emissions remain optimized (Gambini 2023). For the proposed analysis, performances at partial loads and at different ambient temperature of LM9000 in both SCGT, SCGT CHP and CCGT CHP configurations were computed via advanced simulation SW. In each configuration, the GTG is represented as a single-input, multiple-output system that consumes natural gas and generates electricity, heat, and CO₂ emissions.

The considered PV plant is assumed to be a single axis tracking one, with 19% efficiency PV panels, 96% efficiency inverters and a 1.2 ratio DC/AC. Environmental conditions and renewable energy source (RES) output were assessed using the System Advisor Model (SAM) software (Blair 2014). The battery energy storage system to be installed in the hybrid plant is based on a C-rate 1 technology with an overall efficiency of 93% during charging and discharging operations. For the mitigation of fast (intra-hour) PV fluctuations only, see the dedicated paragraph, a C-2 rate BESS again with 93% efficiency was selected.

Optimal design of hybrid power plants

The conventional rules of thumb, design principles, and operational strategies used for fully fossil-fueled energy systems are not suitable for hybrid configurations due to the increased number of components, diverse technologies, complex interactions, and multiple possible layouts (Zelaschi 2016). Therefore, achieving optimal design and operation for hybrid power systems requires the application of advanced mathematical optimization methods.

While several commercial software tools exist for hybrid system design optimization, they are generally tailored to small, grid-connected microgrids primarily powered by diesel generators, and lack the capability to accurately simulate large, off-grid systems using SCGT and CCGT technologies with stringent availability requirements. In collaboration with a leading research center, BH has recently adopted a sophisticated optimization platform (Chong 2013) designed to support the optimal configuration of various hybrid systems. This tool can accurately represent complex Combined Heat and Power (CHP) plants, incorporate constraints to ensure high availability and reliable operation, and model key parameters as functions of environmental conditions (Zelaschi 2016). The optimization goal is to identify the system design that minimizes the Total Annual Cost (TAC), while maintaining energy balance and meeting all reliability and operational constraints, including fault tolerance. TAC includes both operational expenses and capital investment costs, adjusted using the Capital Recovery Function (CRF), and is calculated as: $TAC = OPEX + CAPEX \cdot CRF$. This metric is equivalent to the Levelized Cost of Energy (LCOE), since $LCOE = TAC/YEC$ (Yearly Energy Consumption). The optimization process requires as input: i) thermal and electrical energy demands; ii) renewable energy production and ambient data; iii) performance and cost characteristics of dispatchable units (SCGT/CCGT), battery storage (BESS), and photovoltaic systems (PV). The output includes: i) the selection and sizing of dispatchable, non-dispatchable, and storage units; ii) the operational status and optimal scheduling of these units, including storage systems (e.g., on/off states, start-up/shutdown events, and hourly power set-points).

Mitigation of fast PV fluctuations

A key limitation of the described optimization approach is that it relies on hourly average values for all system variables. As a result, it does not account for short-term variations—such as rapid intra-hour fluctuations in PV output—which may significantly affect system reliability and control strategies. It is important to emphasize that these intra-hour PV power fluctuations, primarily caused by passing clouds, can be extremely severe: ramp rates approaching 5 %/s have been observed in (Cormode 2013; Marcos 2011), while rates exceeding 8 %/s have been reported in (Sukumar 2018) (percentages are relative to the PV plant's nominal power; for a 10 MW_p plant, this corresponds to 500 or 800 kW/s, respectively). Some Transmission System Operators (TSOs) have already addressed the challenges posed by such ramp rates by imposing limits on the allowable power variation rates of grid-connected PV plants, aiming to mitigate their potential impact on the main grid. In response, several strategies have been proposed, typically involving the installation of a dedicated Battery Energy Storage System (BESS), referred to here as BESS_{PV}, operating in parallel with the PV plant solely to smooth out power fluctuations.

The sizing of the BESS_{PV}, in terms of both power and energy capacity, is typically based on a worst-case scenario: a sudden drop in solar irradiance from 100 % to 10 % (diffuse irradiance only) of the maximum annual value recorded at the site (De la Parra 2015). The storage system must be capable of injecting or absorbing sufficient power to ensure that the combined output of the PV plant and BESS_{PV} remains within the permitted ramp rate. However, this ramp-rate-based approach implicitly assumes that, once the ramp ends, the system can always reach a new equilibrium between generation and load—a condition generally valid only in systems with low renewable energy penetration. In off-grid captive power systems with high RES penetration, even if PV fluctuations are smoothed to allow Gas Turbine Generators (GTGs) to ramp accordingly, there may come a point where the balance between generation and demand can no longer be maintained. This occurs when GTGs reach their operational limits and can no longer adjust their output. At that point, the BESS_{PV} must continuously supply or absorb the power imbalance, resulting in a non-zero steady-state power output—even after the PV output has stabilized. The required power profile for the BESS_{PV}, which determines its sizing, hence depends on the available GTGs' power reserve and the PV output level at the start of the fluctuation. Therefore, mitigating fast PV fluctuations using both BESS_{PV} and GTGs requires: i) a high degree of integration in plant control systems; ii) maintaining power reserves in GTGs, which reduces their efficiency; and iii) complex evaluations during the design phase to properly size the BESS_{PV}.

To simplify this scenario, an alternative approach avoids operating GTGs with power reserves to manage PV uncertainty. Instead, the BESS_{PV} alone is used to smooth PV output. In this case, since GTGs output remains unchanged, the BESS_{PV} must fully compensate for the worst-case PV fluctuation. This requires a BESS_{PV} power rating equal to 90 % of the maximum expected PV output for the given location and technology. As for energy capacity, assuming the worst-case scenario where PV output remains constant after the fluctuation, and considering a 30-minute delay to start up offline GTGs if needed, a C2-rate BESS_{PV} is required (note that in case of upward PV fluctuations, which could require the shutoff of some GTGs, turning off some units would be much faster).

Furthermore, since the BESS_{PV} must perform its function even at end-of-life—when its capacity is reduced to approximately 80 % of its original value—we assume that for each MW_p of installed PV, a C2-rate BESS_{PV} is deployed solely to dampen output fluctuations, with a rated power equal to 113 % of the maximum expected PV output for the selected location and technology. The final hybrid power plant configuration is illustrated in Figure 1. The additional CAPEX of BESS_{PV} is added to the cost of PV modules, inverters and electrical infrastructure needed. Since the vast majority of PV fluctuations are well smaller than the worst-case one selected to size the BESS_{PV}, such equipment would be usually cycled with small

Depth-Of-Discharge, and we therefore assume it can last for the whole considered life of the plant (20 years), without need of replacement.

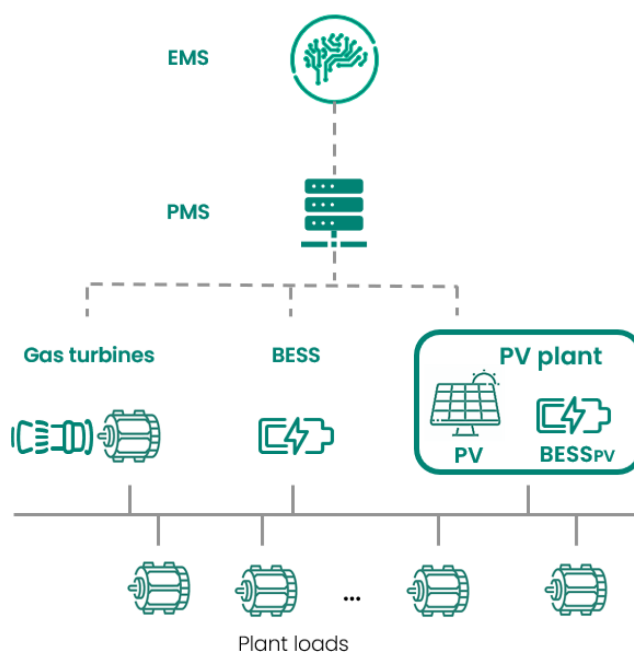


Figure 1—(generated by authors). Hybrid power plant based on GTGs, BESS (for spinning reserve and day/night cycling) and PV plant, the latter including both PV modules with inverters and a dedicated BESS (BESS_{PV}) to smooth out intra-hour PV variations. The picture shows the plant control systems as well, split in EMS (efficiency optimization) and PMS (used e.g. for load dump, frequency regulation, black start).

Decarbonization analysis of a 300 MW e-LNG power generation island

The analyzed scenario involves an LNG facility situated in the Middle East region. The liquefaction process and its associated rotating machinery utilize five NMBL™ liquefaction modules, each powered by electric motor-driven compressors. The liquefaction process demands approximately 300 MW of electrical power and 67 MWth of thermal energy, which also includes estimated loads from LNG plant components not related to NMBL™ (such as pre-treatment and boil-off gas compressors). These energy requirements are met by a dedicated, off-grid power generation system.

The cost of natural gas was fixed at 3 \$/MBTU, representing a typical average for LNG-related uses. To reflect the characteristics of the selected location and to highlight the inherent cost-effectiveness of well-sized hybrid power systems, the analysis excluded both RES incentives and carbon taxes. The capital, installation, and operation and maintenance (O&M) expenses for all technologies evaluated in this study are based on actual current market values, including proprietary data from BH. For the financial modeling, assumptions included a 2% inflation rate, a 20-year expected plant life, and a nominal discount rate of 10%.

All the outcomes are benchmarked against a baseline scenario where all electrical and thermal demands are satisfied using LM9000 SCGT units in both standard and CHP configurations, operated optimally (with an annual average CO₂ intensity of 575 gCO₂/kWh).

Figure 2 illustrates the optimal combination of technologies to be installed for several cases. The (nominal) installed capacity of gas-fired units is based on ISO conditions (15 °C), which means it exceeds the required load because it has to feed the load even under the highest ambient temperatures (up to 47 °C/117 °F in the location under study). The black triangles identify the optimization runs where no constraints on CO₂ emissions are set (the optimization is only devoted to feed the load while minimizing the TAC, installing the best combination of available technologies). The vertical dashed lines represent a separation between different mixes of available technologies. Initially, SCGT only were considered, in both standard

and CHP configuration, to build a reference case for CAPEX, OPEX and carbon dioxide emissions. For this case, only the optimal solution without constraints on emission is provided, because once the load and the GT model are given, the minimization of TAC results in minimizing fuel consumptions, which means minimizing greenhouse gases emissions as well. For the scenarios where i) SCGT and BESS, ii) CCGT and BESS, and iii) SCGT, CCGT and BESS were considered as mix of technologies available for the installation, only the results without CO₂ emissions constraints are given again. In these three cases, however, the reason is that setting decarbonization constraints to i) 10%, ii) 20% and iii) 20%, the optimization problem turned out to be infeasible: even in these scenarios, therefore, the solution at minimum total annual cost, which provides a decarbonization of i) 7.6%, ii) 16.8% and iii) 17.0%, is very close to the maximum decarbonization achievable. From Figure 2, we see that just the installation of a BESS can decrease nearly by 8% the greenhouse gases emissions of the plant, since the spinning reserve is provided by the battery and the GTGs can run at higher partial loads characterized by higher efficiencies. An upgrade of some SCGT to CCGT can lead to reach about 17% of decarbonization, thanks to the increase in the efficiency of the generators (the result in terms of decarbonization is close in both configurations with or without the BESS, since adding a BESS the optimizer obtains a lower TAC keeping a greater portion of the system in SCGT configuration). When PV is added to the mix of available technologies that can be installed, on the other hand, it is always possible to find higher decarbonizing solutions, going well above the 21% guaranteed by the minimum TAC hybrid power plant. It is evident that the optimal configurations involve a balanced integration of all available technologies (LM9000 SC, LM9000 SC CHP, LM9000 CC CHP, BESS, and PV), with varying proportions depending on the decarbonization target relative to simple cycle gas turbines: as the decarbonization increases, the size of PV and BESS systems clearly expands. Notably, there is a range (21% - 35%) where the optimal mix of GT-based technologies and BESS remains constant, while only the PV plant size grows significantly with decarbonization. This feature can be strategically used to achieve the final decarbonization target through phased investments, each aimed at increasing PV capacity and thereby lowering initial capital expenditure. We highlight that a portion of SCGT units is always present, since they are suitable for daily cycling, making them effective in balancing PV power fluctuations. For the sake of clarity, Figure 2 shows the size of PV without explicitly mentioning BESS_{PV}, but, as per the analysis illustrated in the previous paragraph, for each MWp of PV plant, it is assumed that a 0.94 MW C2-rate BESS_{PV} is installed in parallel, and its costs are considered into the CAPEX and OPEX of the whole PV plant.

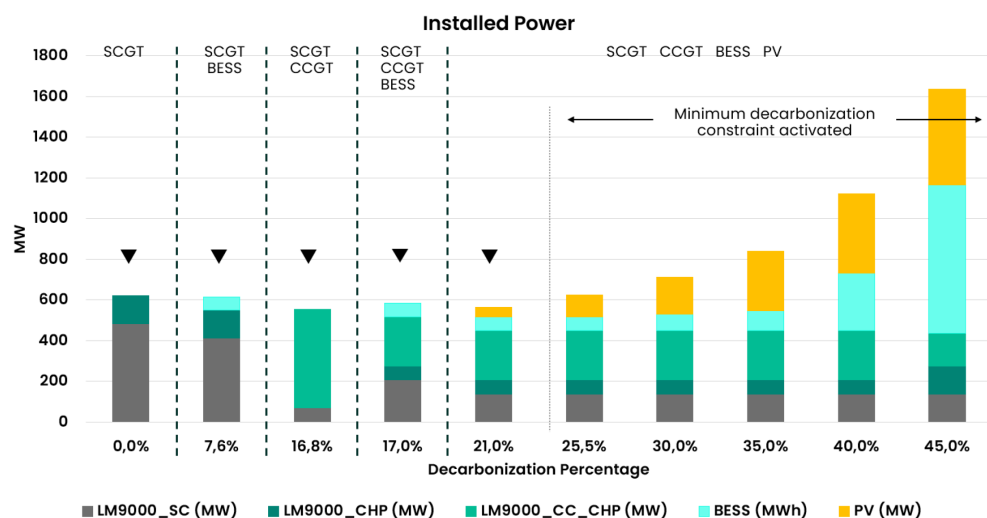


Figure 2—(generated by authors). Optimal mix of technologies to be installed as a function of decarbonization percentage. Black triangles: solutions without constraints on minimum decarbonization target. Vertical dashed lines: separation between different mix of technologies available for installation.

Figure 3 and Figure 4 display the Capital Expenditures (CAPEX) and Operational Expenditures (OPEX) of the optimal hybrid configurations, expressed in per unit (p.u.) relative to the reference scenario. Plants with SCGT, CCGT and BESS are characterized by CAPEX increases of less than 20% with respect to the reference one. The same holds for the hybrid power plant (SCGT, CCGT, BESS and PV) without CO₂ emissions constraint, then CAPEX rises gradually up to a 35% decarbonization level, primarily due to the expansion of the PV plant. Within this range, the optimal configurations include a relatively small BESS (~70 - 100 MWh), which serves mainly to provide spinning reserve, as renewable energy is consumed only when it is generated. This operational strategy allows the GTGs to reduce their spinning reserve requirements compared to the reference case, thereby operating more efficiently. Beyond the 35% decarbonization threshold, a noticeable shift in the CAPEX trend occurs, driven by a rapid increase in both PV and BESS capacities: for instance, achieving 40% CO₂ reduction requires a ~400 MWp PV system, with its 384 MWh BESS_{PV}, and a ~300 MWh BESS that is cycled daily. The key factor behind this sharp CAPEX increase is the need to utilize PV energy at night, which necessitates storing daytime solar energy. This, in turn, requires a large-scale BESS, which undergoes daily charge/discharge cycles and is therefore subject to degradation and costly replacement after a few years. The impact of BESS wear is reflected in the OPEX trend, which, for all the mix of technologies, declines up to 35% decarbonization due to fuel savings, but then begins to rise at higher decarbonization levels as battery degradation from cycling offsets the reduction in fuel use.

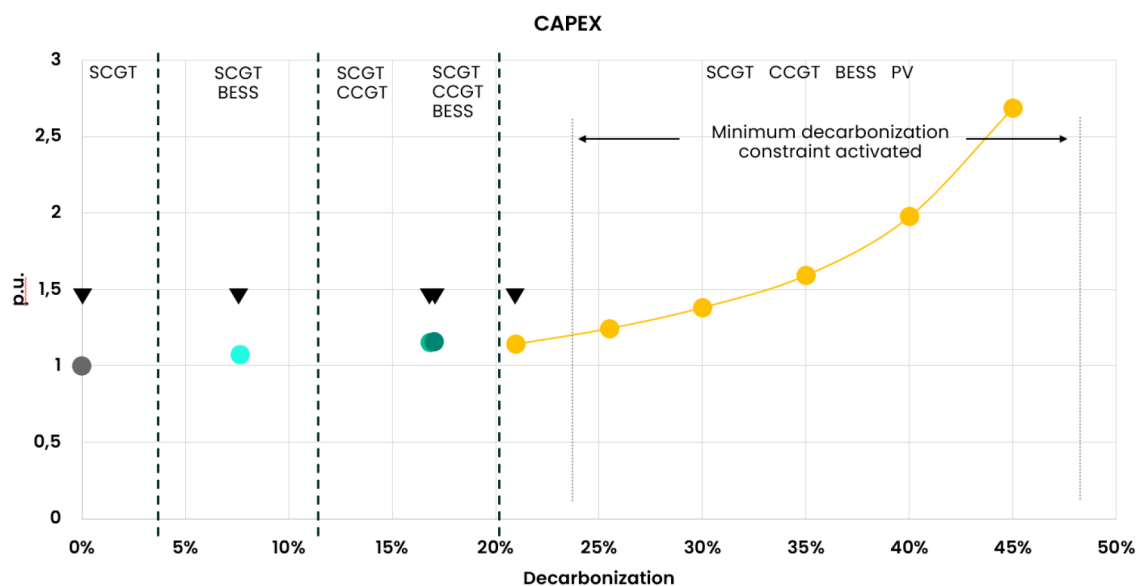


Figure 3—(generated by authors). CAPEX, in p.u. with respect to the reference case with SCGT only, of the optimal mixes of dispatchable and non-dispatchable units to be installed. Black triangles: solutions without constraints on minimum decarbonization target. Vertical dashed lines: separation between different mix of technologies available for installation.



Figure 4—(generated by authors). OPEX, in p.u. with respect to the reference case with SCGT only, of the optimal mixes of dispatchable and non-dispatchable units to be installed. Black triangles: solutions without constraints on minimum decarbonization target. Vertical dashed lines: separation between different mix of technologies available for installation

These CAPEX and OPEX patterns result in the Total Annualized Cost (TAC) shown in Figure 5, normalized against the reference case. The TAC reaches its lowest point at around 20% carbon reduction, showing a decrease of approximately 10%. The normalized TAC of the hybrid power plants remains below 1 up to 30% decarbonization and matches the reference case between 30% and 35% carbon reduction. This defines the upper boundary of the range where hybrid power plants with decarbonization targets are still economically favorable compared to the reference plant. Notice that, as long as the normalized TAC is under 1, hybrid solutions reduce both emissions and costs, resulting in a negative carbon abatement cost.

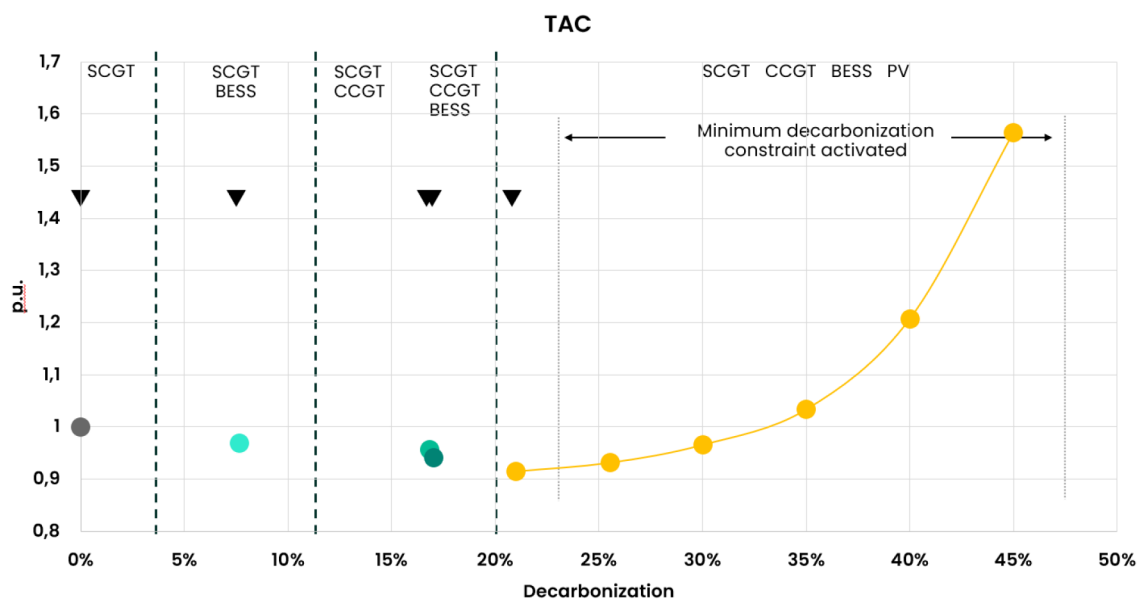


Figure 5—(generated by authors). TAC, in p.u. with respect to the reference case with SCGT only, of the optimal mixes of dispatchable and non-dispatchable units to be installed. Black triangles: solutions without constraints on minimum decarbonization target. Vertical dashed lines: separation between different mix of technologies available for installation

For the sake of completeness, we provide a table summarizing the main figures described above. The discounted payback time was computed assuming a 2% inflation, a 9.9% nominal discount rate and 20 years of plant lifetime

Table 1—Summary of installed technology mix and main economic figures for the selected use case.

	SCGT (min TAC)	SCGT, BESS (min TAC)	SCGT, CCGT (min TAC)*	SCGT, CCGT, BESS (min TAC)*	SCGT, CCGT, BESS, PV (min TAC)	SCGT, CCGT, BESS, PV (CO2: -25%)	SCGT, CCGT, BESS, PV (CO2: -30%)	SCGT, CCGT, BESS, PV (CO2: -35%)	SCGT, CCGT, BESS, PV (CO2: -40%)	SCGT, CCGT, BESS, PV (CO2: -45%)
SCGT (#)	7	6	1	3	2	2	2	2	2	2
CCGT(#)	0	0	6	3	3	3	3	3	3	2
BESS (MWh)	0	62,4	0	67,8	68,2	67,8	82,8	99,5	282,3	729
PV (MWp)	0	0	0		48,6	109,4	179,2	291	390,7	470,6
Carbon intensity [gCO ₂ /kWh]	574,7	530,8	477,9	476,7	454,1	428,0	402,2	373,5	344,8	316,1
Decarb. percentage	0	7,6	16,8	17,0	20,9	25,5	30,0	35,0	40,0	45,0
CAPEX (p.u.)	1	1,07	1,15	1,15	1,13	1,24	1,37	1,58	1,97	2,68
OPEX (p.u.)	1	0,90	0,83	0,81	0,78	0,74	0,72	0,70	0,75	0,89
TAC (p.u.)	1	0,96	0,95	0,94	0,91	0,93	0,96	1,03	1,20	1,56
LCOE (p.u.)	1	0,97	0,95	0,94	0,91	0,93	0,97	1,03	1,20	1,55
Discounted paybak time (years)	-	6	8	7	5	8	13	>20	>20	>20

Figure 6 illustrates the operation of the optimal hybrid power plant that achieves a 30% reduction in CO₂ emissions, shown for a representative day that effectively captures system behavior under varying ambient conditions. The figure presents cumulative electric power balance graphs, where the electric demand of the e-LNG facility is equal to 300 MW. The optimal configuration features a baseload supplied by the CCGT LM9000, which offers the highest efficiency among the available gas turbines, and a SCGT CHP LM9000 unit to meet thermal energy requirements. During midday hours, a noticeable portion of electricity is generated by the PV system, while the SCGT LM9000 units operate at low partial loads or are temporarily shut down. At night, the plant is powered solely by gas turbines operating in simple or combined cycle mode (SCGT and CCGT). The battery energy storage system primarily functions as a power reserve, providing spinning reserve in case of failure, which enables the gas turbines to avoid operating at highly inefficient load points.

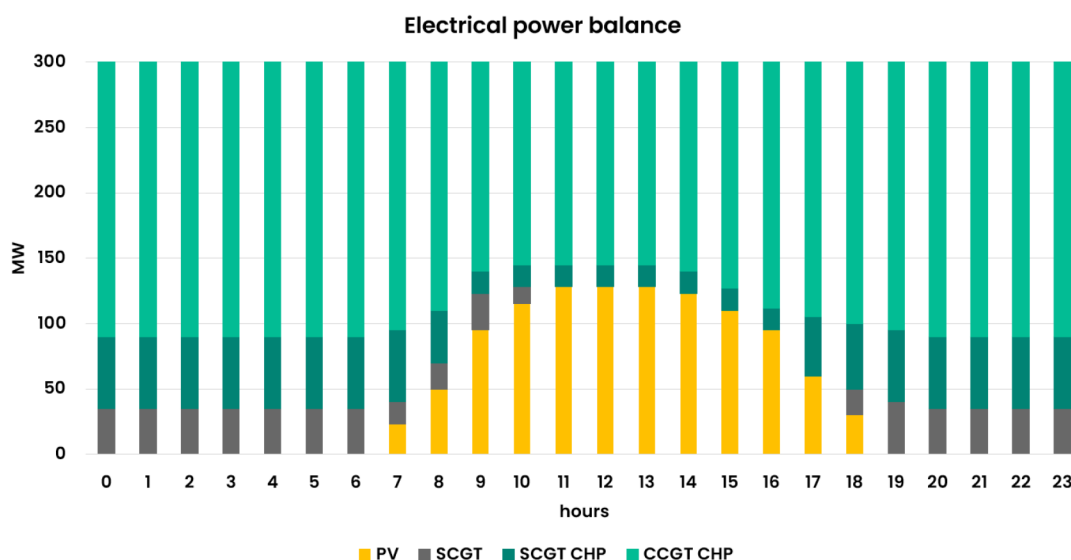


Figure 6—(generated by authors). Energy balance of the considered LNG plant during a typical day of operation. Vertical bars correspond to the power produced at each hour by each power source

Conclusions and Recommendations

This study focuses on the optimal design of a hybrid captive power plant comprising gas turbines (both SCGT and CCGT), battery energy storage systems (BESS), and photovoltaic (PV) units, tailored for a representative e-LNG application with stringent availability requirements. The problem is formulated as a cost-minimization optimization model, targeting the reduction of total annual cost (TAC), which encompasses both capital expenditures and operational expenses. The model integrates comprehensive availability constraints and enables optimal capacity sizing while adhering to predefined decarbonization objectives. The optimization process evaluates the generation mix and system performance under progressively stringent decarbonization targets. Each one of the considered technologies has been inserted separately, so to clearly highlight its contribution to the decarbonization, to the CAPEX and to the OPEX. Results are benchmarked against a reference configuration relying solely on simple cycle gas turbines (SCGT). The findings reveal that, even in the absence of subsidies or carbon pricing mechanisms, and under current realistic cost assumptions, a properly optimized combination of conventional and renewable technologies with storage can achieve decarbonization while reducing overall costs. Specifically, the analysis shows that up to approximately 35% decarbonization can be achieved without a significant increase in TAC, while maintaining high system robustness. In the examined case, the cost-optimal hybrid configuration achieves over 20% decarbonization and reduces TAC by around 8% compared to the SCGT baseline. Furthermore, a 30% reduction in emissions is attainable at no additional cost, and a 35% reduction is feasible with only a marginal TAC increase (~5%). Notably, the optimal configuration of GTGs and BESS remains consistent across a broad decarbonization range, supporting the feasibility of a phased investment strategy toward long-term decarbonization goals.

Nomenclature

BESS	Battery Energy Storage System
BH	Baker Hughes
CAPEX	Capital Expenditures
CC	Combined Cycle
CCGT	Combined Cycle Gas Turbine
CHP	Combined Heat and Power
CRF	Capital Recovery Factor

e-LNG	Electrified Liquefied Natural Gas
EMS	Energy Management System
GHG	Greenhouse Gas
GT	Gas Turbine
GTG	Gas Turbine Generator
LCCA	Levelized Cost of Carbon Abatement
LCOE	Levelized Cost Of Energy
LNG	Liquefied Natural Gas
MILP	Mixed-Integer Linear Programming
MTPA	Million Tonnes Per Annum
MW	MegaWatt
MWh	MegaWatt-hour
MWp	MegaWatt peak
O&M	Operation & Maintenance
OPEX	Operational Expenditure
PMS	Power Management System
p.u.	per unit
PV	Photovoltaic
RES	Renewable Energy Sources
SAM	System Advisor Model
SC	Simple Cycle
SCGT	Simple Cycle Gas Turbine
SOC	State Of Charge
ST	Steam Turbine
TAC	Total Annual Cost
TSO	Transmission System Operator
YEC	Yearly Energy Consumption

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